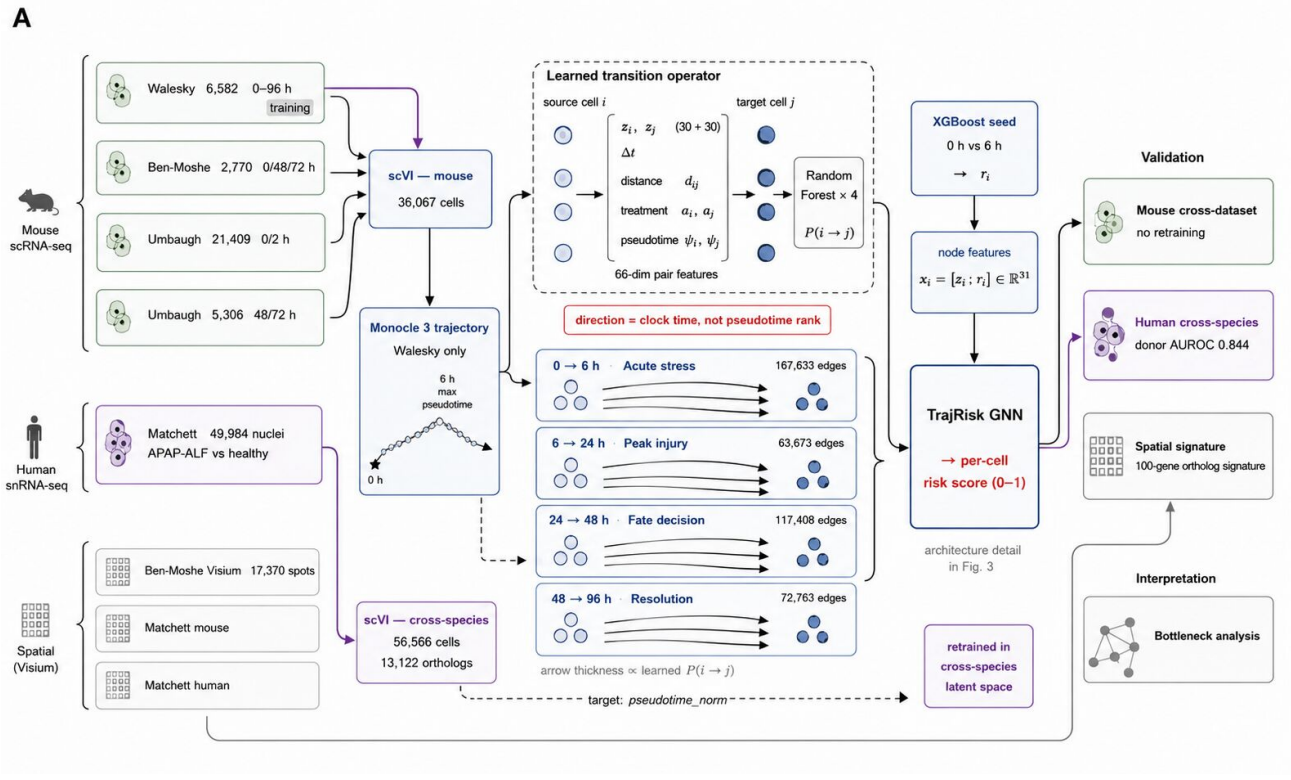
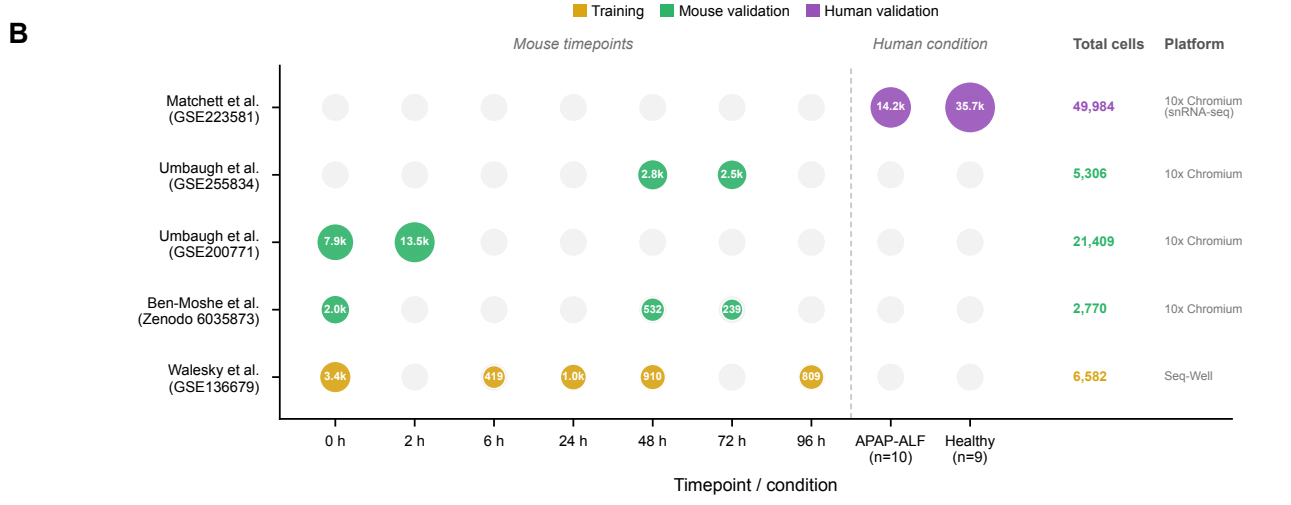
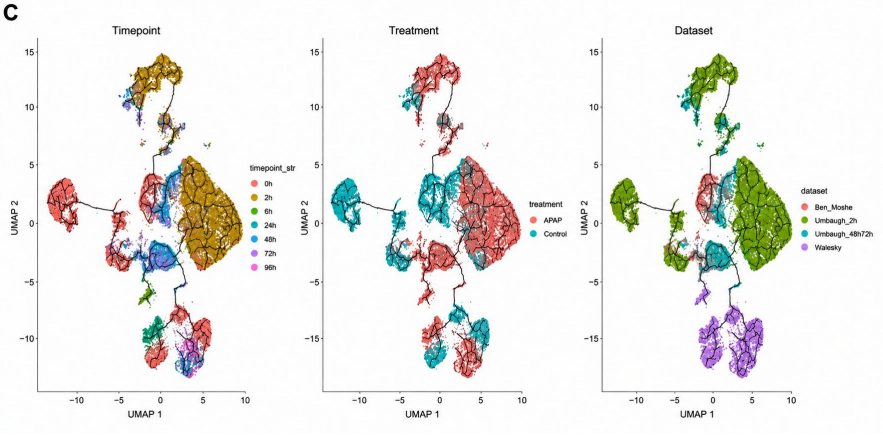
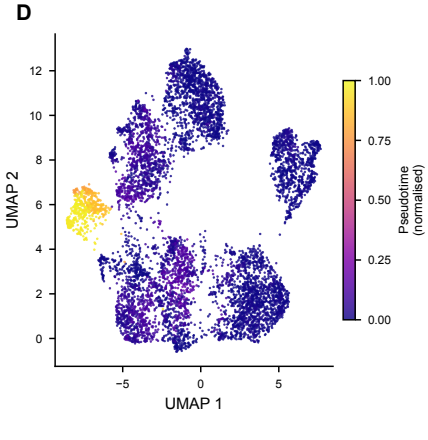

A
Mouse scRNA-seq
Walesky 6,582 0–96 h
training
Ben-Moshe 2,770 0/48/72 h
Umbaugh 21,409 0/2 h
Umbaugh 5,306 48/72 h
scVI — mouse
36,067 cells
Monocle 3 trajectory
Walesky only
6 h
max
pseudotime
0 h
Human snRNA-seq
Matchett 49,984 nuclei
APAP-ALF vs healthy
Spatial (Visium)
Ben-Moshe Visium 17,370 spots
Matchett mouse
Matchett human
scVI — cross-species
56,566 cells
13,122 orthologs
Learned transition operator
source cell i
target cell j
z_i, z_j (30 + 30)
Δt
distance d_{ij}
treatment a_i, a_j
pseudotime ψ_i, ψ_j
66-dim pair features
Random Forest × 4
$P(i \to j)$
direction = clock time, not pseudotime rank
0 → 6 h · Acute stress 167,633 edges
6 → 24 h · Peak injury 63,673 edges
24 → 48 h · Fate decision 117,408 edges
48 → 96 h · Resolution 72,763 edges
arrow thickness ∝ learned $P(i \to j)$
target: pseudotime_norm
XGBoost seed
0 h vs 6 h
→ r_i
node features
$x_i = [z_i ; r_i] \in \mathbb{R}^{31}$
TrajRisk GNN
→ per-cell risk score (0–1)
architecture detail in Fig. 3
retrained in cross-species latent space
Validation
Mouse cross-dataset
no retraining
Human cross-species
donor AUROC 0.844
Spatial signature
100-gene ortholog signature
Interpretation
Bottleneck analysis


B
Training
Mouse validation
Human validation
Mouse timepoints
Human condition
Total cells
Platform
Matchett et al. (GSE223581)
14.2k
35.7k
49,984
10x Chromium (snRNA-seq)
Umbaugh et al. (GSE255834)
2.8k
2.5k
5,306
10x Chromium
Umbaugh et al. (GSE200771)
7.9k
13.5k
21,409
10x Chromium
Ben-Moshe et al. (Zenodo 6035873)
2.0k
532
239
2,770
10x Chromium
Walesky et al. (GSE136679)
3.4k
419
1.0k
910
809
6,582
Seq-Well
0 h
2 h
6 h
24 h
48 h
72 h
96 h
APAP-ALF (n=10)
Healthy (n=9)
Timepoint / condition


C
Timepoint
Treatment
Dataset
UMAP 1
UMAP 2
timepoint_str
0h
2h
6h
24h
48h
72h
96h
treatment
APAP
Control
dataset
Ben_Moshe
Umbaugh_2h
Umbaugh_48h72h
Walesky


D
UMAP 1
UMAP 2
Pseudotime (normalised)
1.00
0.75
0.50
0.25
0.00